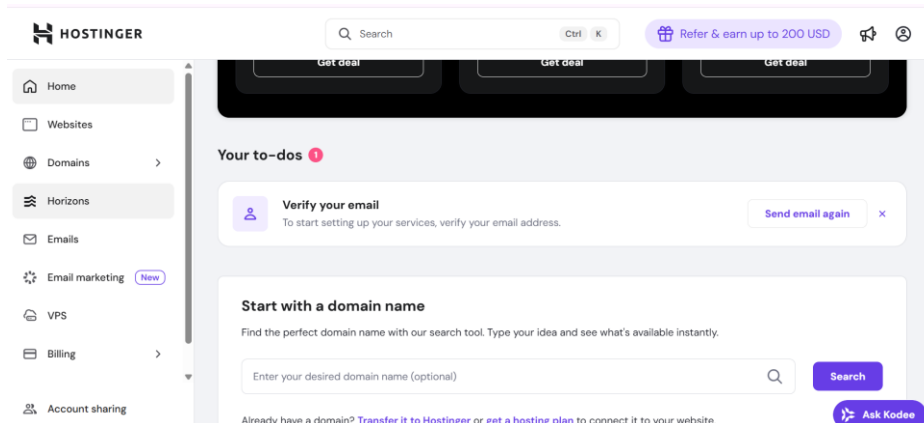


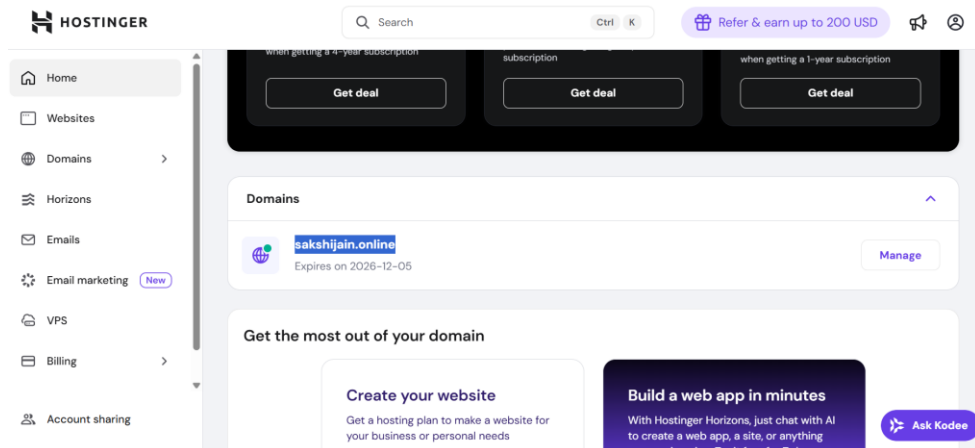
Route 53 – Web Hosting using ALB

Route 53 converts domain names (like google.com) into IP addresses (like 142.250.192.78) so users can reach websites and applications.

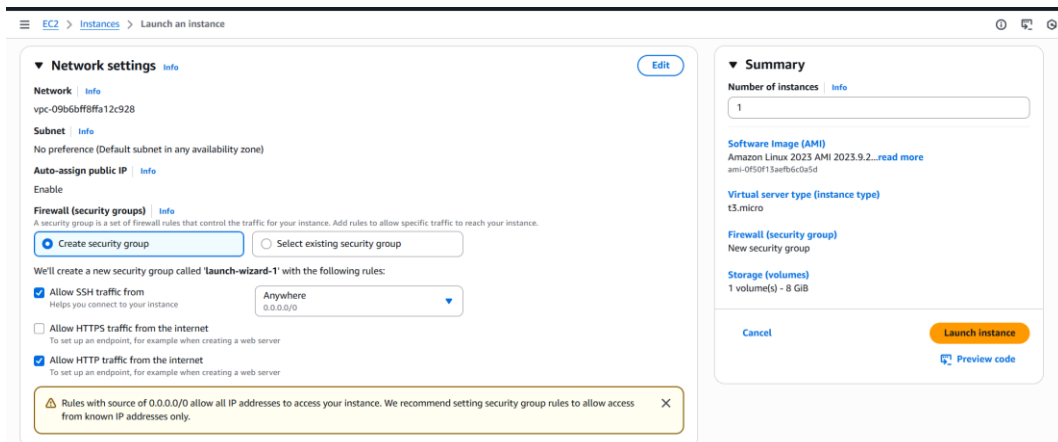
1. First you Need to create account on hostinger.com



2. Purchase Domain Name



3. Launch EC2 Instances For Home, Contact



2

Allow tags in metadata [Info](#)

Select

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<h1>This is my HomePage of $HOSTNAME</h1>" > /var/www/html/index.html
```

☐ User data has already been base64 encoded

Summary

Number of instances [Info](#)

1

When launching more than 1 instance, consider EC2 Auto Scaling

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)

ami-0f50f13aefb6c0a5d

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

2

Allow tags in metadata [Info](#)

Select

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum install httpd -y
systemctl start httpd
systemctl enable httpd
mkdir -p /var/www/html/contact
echo "<h1>This is my contact page of $HOSTNAME</h1>" > /var/www/html/contact/index.html
```

☐ User data has already been base64 encoded

Summary

Number of instances [Info](#)

2

When launching more than 1 instance, consider EC2 Auto Scaling

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)

ami-0f50f13aefb6c0a5d

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Instances (4) [Info](#)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☐	Contact	i-0e89f27456b208c31	Running	t3.micro	Initializing	View alarms +	eu-north-1a	ec2-13-49-46-52.eu-no
☐	Contact	i-0d049598ce2220208	Running	t3.micro	Initializing	View alarms +	eu-north-1a	ec2-16-171-239-82.eu-
☐	Home	i-06126d0d7e89b4103	Running	t3.micro	3/3 checks pass	View alarms +	eu-north-1a	ec2-16-16-213-187.eu-
☐	Home	i-0f13bd1e40d9568ac	Running	t3.micro	3/3 checks pass	View alarms +	eu-north-1a	ec2-13-61-5-162.eu-no

4. Create Security Group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Application-load-balancer

Name cannot be edited after creation.

Description [Info](#)

security group

VPC [Info](#)

vpc-09b6bfff8fa12c928

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	
HTTP	TCP	80	Anywh...	0.0.0.0/0	Delete
SSH	TCP	22	Anywh...	0.0.0.0/0	Delete

[Add rule](#)

5. Create Target Group

EC2 > Target groups > Create target group

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name
Name must be unique per Region per AWS account.
Home-target
Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 11/32

Protocol **Port**

Click on include as pending below

EC2 > Target groups > Create target group

Register targets - recommended
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/4)

Filter instances

Instance ID	Name	State	Security groups	Zone
☐ i-0e89f27456b208c31	Contact	Running	launch-wizard-3	eu-north-1a
☐ i-0d049598ce2220208	Contact	Running	launch-wizard-3	eu-north-1a
☒ i-06126d8d7e89b4103	Home	Running	launch-wizard-2	eu-north-1a
☒ i-0f13bd1e40d9568ac	Home	Running	launch-wizard-2	eu-north-1a

2 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.
80
1-65535 (separate multiple ports with comma)

[Include as pending below](#)

Do the same process for Contact Instance

Target groups (2) [Info](#) [What's new?](#)

Filter target groups

	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
☐	Contact-Target	arn:aws:elasticloadbalancin...	80	HTTP	Instance	None associated	vpc-09b6bff8ffa12c9
☐	Home-target	arn:aws:elasticloadbalancin...	80	HTTP	Instance	None associated	vpc-09b6bff8ffa12c9

6. Create Application Load Balancer

EC2 > Load balancers > Create Application Load Balancer

Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When traffic request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

How Application Load Balancers work

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
ALP
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**

EC2 > Load balancers > Create Application Load Balancer

vpc-09b6bfff8fa12c928 (default) Create VPC

IP pools [Info](#)
 You can optionally choose to configure an IPAM pool as the preferred source for your load balancer IP addresses. Create or view pools in the [Amazon VPC IP Address Manager console](#).
☐ Use IPAM pool for public IPv4 addresses
 The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)
 Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ eu-north-1a (eu-n1-az1)
 Subnet
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
 subnet-0791f94763c174151
 IPv4 subnet CIDR: 172.31.16.0/20

☒ eu-north-1b (eu-n1-az2)
 Subnet
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
 subnet-03c9ded9fc50afdc
 IPv4 subnet CIDR: 172.31.32.0/20

☒ eu-north-1c (eu-n1-az3)
 Subnet
 Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.
 subnet-04712f4123f4a8913
 IPv4 subnet CIDR: 172.31.0.0/20

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

Application-load-balancer
 sg-0cc0b31422f811d45 VPC: vpc-09b6bfff8fa12c928

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 Remove

Protocol HTTP Port 80
 1-65535

Default action [Info](#)
 The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action
☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed response

Forward to target group [Info](#)
 Choose a target group and specify routing weight or [create target group](#).

Target group
 Home-target HTTP
 Target type: Instance, IPv4 | Target stickiness: Off

Weight 1
 0-999

Percent 100%

Goto Listener and Rules an add rule for Contact Page

Listeners and rules Network mapping Resource map Security Monitoring Integrations Attributes Capacity Tags

Listeners and rules (1/1) [Info](#) Manage rules Manage listener Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the rules you define and any additional rules.

Filter listeners

Protocol:Port	Default action	Rules	ARN	Default SSL/TLS certificate
HTTP:80	Forward to target group Home-target (1 (100%)) Target group stickiness: Off	1 rule	ARN	Not applicable

EC2 > Load balancers > ALP > HTTP:80 listener > Add rule

Add rule [Info](#)
 Requests that reach this rule are evaluated against its conditions. If a request matches all of the rule conditions, then a secondary evaluation is done to perform any specified transforms, and finally the request is routed according to the rule's actions.

Step 1: Add rule
 Step 2: Set rule priority
 Step 3: Review and create

► **Listener details:** HTTP:80

► **Name and tags** [Info](#)
 Tags can help you manage, identify, organize, search for and filter resources.

Conditions (1 value) [Info](#) [Rule limits](#)
 Define 1-5 condition values. Additional conditions can't be added once the limit is reached.

▼ **Path (value)** = /contact/* Remove

Match pattern type
☒ Value matching Match with glob syntax, using * and ? as wildcards.
☐ Regex matching Match with regex syntax.

Path condition value
 Case sensitive.
 = /contact/*
 Valid characters are a-z, 0-9, /, ., -, and special characters. Paths must be 1-128 characters.

Actions [Info](#)
Requests matching all rule conditions route according to the rule actions.

Routing action

☒ Forward to target groups
 ☐ Redirect to URL
 ☐ Return fixed response

Forward to target group [Info](#)
Choose a target group and specify routing weight or [create target group](#).

Target group
 Contact-Target
 Target type: Instance, IPv4 | Target stickiness: Off

Weight
 1
 0-999

Percent
 100%

+ Add target group
You can add up to 4 more target groups.

Target group stickiness [Info](#)
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

[Cancel](#)
[Skip to Review changes](#)
[Next](#)

EC2 > Load balancers > ALP > HTTP:80 listener > Add rule

Step 1
Add rule

Step 2
Set rule priority

Step 3
Review and create

Set rule priority [Info](#)
Each rule has a priority. The default rule is evaluated last. You can change the priority of a non-default rule at any time. You can't change the priority of the default rule.

► Listener details: HTTP:80

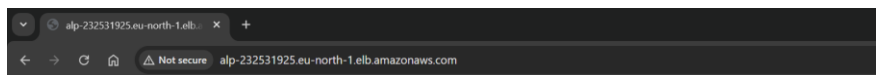
Listener rules (2) [Info](#) [Rule limits](#) [Reset priorities](#) [Add gap between priorities](#)

Traffic received by the listener is routed according to the default action and any additional rules. Rules are evaluated in priority order from the lowest value to the highest value.

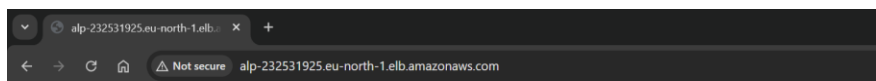
Priority	Name tag	Conditions (If)	Transforms	Actions (Then)
1 Priority value must be 1-50,000.	-	Path (value) = /contact/*	-	Forward to target group Home-target: 1 (100%) Target group stickiness: Off
Last (default)	Default	If no other rule applies	-	Forward to target group Home-target: 1 (100%) Target group stickiness: Off

[Cancel](#)
[Previous](#)
[Next](#)

Run DNS of Load Balancer You will get your webpage
For Home Page

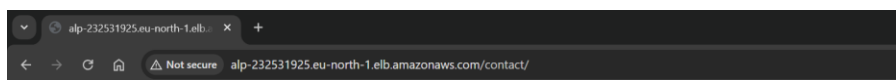


This is my HomePage of ip-172-31-24-194.eu-north-1.compute.internal

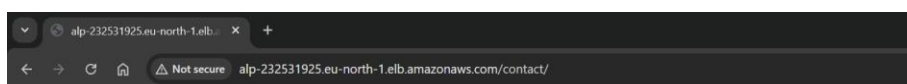


This is my HomePage of ip-172-31-24-194.eu-north-1.compute.internal

For Contact Page



This is my contact page of ip-172-31-22-212.eu-north-1.compute.internal



This is my contact page of ip-172-31-27-184.eu-north-1.compute.internal

7. Create Hosted zone

The screenshot shows the 'Create hosted zone' page in the AWS Management Console. The page is titled 'Create hosted zone' and includes a 'Hosted zone configuration' section. The 'Domain name' field is set to 'sakshijain.online'. The 'Description' field is set to 'Sample website'. The 'Type' is set to 'Public hosted zone'. A success message at the top states: 'Record for sakshijain.online was successfully created. Route 53 propagates your changes to all of the Route 53 authoritative DNS servers within 60 seconds. Use "View status" button to check propagation status.'

Hosted zone configuration
A hosted zone is a container that holds information about how you want to route traffic for a domain, such as example.com, and its subdomains.

Domain name [Info](#)
This is the name of the domain that you want to route traffic for.

Valid characters: a-z, 0-9, !, #, \$, %, &, ' () * +, -, . / : ; < > ? @ [\] ^ _ ` { | } . ~

Description - optional [Info](#)
This value lets you distinguish hosted zones that have the same name.

The description can have up to 256 characters. 14/256

Type [Info](#)
The type indicates whether you want to route traffic on the internet or in an Amazon VPC.

☒ **Public hosted zone**
A public hosted zone determines how traffic is routed on the internet.

☐ **Private hosted zone**
A private hosted zone determines how traffic is routed within an Amazon VPC.

8. Copy Nameserver in Hostinger

The screenshot shows the 'DNS / Nameservers' page in the Hostinger control panel. The page displays a list of nameservers: 'ns1.dns-parking.com' and 'ns2.dns-parking.com'. Below this, there is a section titled 'Select Nameservers' with two options: 'Use Hostinger nameservers (recommended)' and 'Change nameservers'. The 'Change nameservers' option is selected. Below this, there are four input fields for the nameservers, with the first three containing 'ns-1143.awsdns-14.org', 'ns-419.awsdns-52.com', and 'ns-883.awsdns-46.net'. The fourth field is empty. At the bottom, there are 'Save' and 'Cancel' buttons, and a 'Ask Kodee' button.

HOSTINGER [Search](#) [Refer & earn up to 200 USD](#)

Domain Overview
DNS / Nameservers
Domain Ownership
Brand protection

Select Nameservers
☐ Use Hostinger nameservers (recommended)
☒ Change nameservers

[Save](#) [Cancel](#) [Ask Kodee](#)

9. Create New Record

Create record

Quick create record

Record 1

Record name: subdomain sakshijain.online

Record type: A – Routes traffic to an IPv4 address and some AWS resources

Route traffic to: Alias to Application and Classic Load Balancer

Europe (Stockholm)

Value: dualstack.ALP-1331791062.eu-north-1.elb.amazonaws.com

Routing policy: Simple routing

Evaluate target health: Yes

Copy Record name and search it.

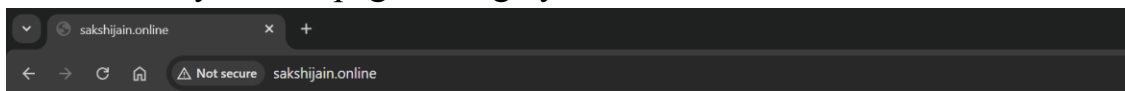
Route 53

Hosted zones

Records (1/3)

Record	Type	Routing policy	Difference	Alias	Value
sakshijain...	A	Simple	-	Yes	dualsta

10. You can see your webpage through your domain name



This is my HomePage of ip-172-31-21-147.eu-north-1.compute.internal



This is my contact page of ip-172-31-22-212.eu-north-1.compute.internal